Econometría

8 | Regresión simple

Supuestos y bondad de ajuste

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OLS

Supuestos MCRL (Gauss)

1. Linealidad en los parámetros.
2. Regresor fijo.
3. $E(u_i|x_{i2}) = E(u_i) = 0 \quad$ (por S2).
4. Homoscedasticidad: $var(u_i|x_{i2}) = \sigma^2 \Rightarrow var(y_i|x_{i2}) = \sigma^2$.
5. No autocorrelación: $cov(u_i, u_j|x_{i2}, x_{j2}) = 0, i \neq j$.
6. N grande.
7. No *outliers*.

Precisión LSE

La varianza y error estándar (ee) de los LSE son:

$$var(\hat{\beta}_2) = \frac{\sigma^2}{\sum(x_{i2} - \bar{x}_2)^2} \tag{1}$$

$$ee(\hat{\beta}_2) = \frac{\sigma}{\sqrt{\sum(x_{i2} - \bar{x}_2)^2}} \tag{2}$$

$$var(\hat{\beta}_1) = \frac{\sum x_{i2}^2}{n\sum(x_{i2} - \bar{x}_2)^2}\,\sigma^2 \tag{3}$$

$$ee(\hat{\beta}_1) = \sigma\sqrt{\frac{\sum x_{i2}^2}{n\sum(x_{i2} - \bar{x}_2)^2}} \tag{4}$$

Precisión LSE

Se sabe que $y_i - \bar{y} = \beta_2(x_{i2} - \bar{x}_2) + (u_i - \bar{u})$ y, además, $\sum(x_{i2} - \bar{x}_2) = \sum \underline{x}_{i2} = 0$. Luego,

$$\begin{aligned}
\hat{\beta}_2 &= \frac{\sum(x_{i2} - \bar{x}_2)(y_i - \bar{y})}{\sum(x_{i2} - \bar{x}_2)^2} \\
&= \frac{\sum(x_{i2} - \bar{x}_2)[\beta_2(x_{i2} - \bar{x}_2) + (u_i - \bar{u})]}{\sum(x_{i2} - \bar{x}_2)^2} \\
&= \frac{\beta_2 \sum(x_{i2} - \bar{x}_2)^2 + \sum(x_{i2} - \bar{x}_2)(u_i - \bar{u})}{\sum(x_{i2} - \bar{x}_2)^2} \\
&= \beta_2 + \frac{\sum(x_{i2} - \bar{x}_2)u_i}{\sum(x_{i2} - \bar{x}_2)^2} \qquad (5)
\end{aligned}$$

Precisión LSE

$$
\begin{aligned}
var(\hat{\beta}_2) &= var\left(\frac{\sum(x_{i2}-\bar{x}_2)u_i}{\sum(x_{i2}-\bar{x}_2)^2}\right) \\
&= \frac{1}{\left[\sum(x_{i2}-\bar{x}_2)^2\right]^2} var\left(\sum(x_{i2}-\bar{x}_2)u_i\right)
\end{aligned}
$$

$$
var\left(\sum(x_{i2}-\bar{x}_2)u_i\right) = \sum(x_{i2}-\bar{x}_2)^2 var(u_i) = \sigma^2 \sum(x_{i2}-\bar{x}_2)^2
$$

$$
\begin{aligned}
var(\hat{\beta}_2) &= \frac{\sigma^2 \sum(x_{i2}-\bar{x}_2)^2}{\left[\sum(x_{i2}-\bar{x}_2)^2\right]^2} \\
&= \frac{\sigma^2}{\sum(x_{i2}-\bar{x}_2)^2}
\end{aligned}
$$

Estimador de la varianza

$$
\begin{aligned}
\hat{u}_i &= y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} \\
&= u_i + \beta_1 - \hat{\beta}_1 + (\beta_2 - \hat{\beta}_2) x_{i2} \\
&= u_i + \beta_1 - \bar{y} + \hat{\beta}_2 \bar{x}_2 + (\beta_2 - \hat{\beta}_2) x_{i2} \\
&= u_i + \beta_1 - (\beta_1 + \beta_2 \bar{x}_2 + \bar{u}) + \hat{\beta}_2 \bar{x}_2 + (\beta_2 - \hat{\beta}_2) x_{i2} \\
&= u_i - \bar{u} + (\hat{\beta}_2 - \beta_2) \bar{x}_2 + (\beta_2 - \hat{\beta}_2) x_{i2} \\
&= (u_i - \bar{u}) - (\hat{\beta}_2 - \beta_2)(x_{i2} - \bar{x}_2) \\
\sum \hat{u}_i^2 &= \sum \left[(u_i - \bar{u}) - (\hat{\beta}_2 - \beta_2)(x_{i2} - \bar{x}_2) \right]^2 \\
&= \sum (u_i - \bar{u})^2 - 2(\hat{\beta}_2 - \beta_2) \sum (x_{i2} - \bar{x}_2)(u_i - \bar{u}) + (\hat{\beta}_2 - \beta_2)^2 \sum (x_{i2} - \bar{x}_2)^2
\end{aligned}
$$

Estimador de la varianza

$$\sum(x_{i2} - \bar{x}_2)(u_i - \bar{u}) = \sum(x_{i2} - \bar{x}_2)u_i \quad [\text{de } \sum(x_i - \bar{x}) = 0]$$

$$\hat{\beta}_2 - \beta_2 = \frac{\sum(x_{i2} - \bar{x}_2)u_i}{\sum(x_{i2} - \bar{x}_2)^2} \quad [\text{de } (5)]$$

$$\sum \hat{u}_i^2 = \sum(u_i - \bar{u})^2 - \frac{\left[\sum(x_{i2} - \bar{x}_2)u_i\right]^2}{\sum(x_{i2} - \bar{x}_2)^2}$$

$$\mathbb{E}\left[\sum(u_i - \bar{u})^2\right] = (n-1)\sigma^2; \quad \mathbb{E}\left[\frac{\left(\sum(x_{i2} - \bar{x}_2)u_i\right)^2}{\sum(x_{i2} - \bar{x}_2)^2}\right] = \sigma^2$$

$$\mathbb{E}\left(\sum \hat{u}_i^2\right) = (n-2)\sigma^2$$

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2} \quad \Rightarrow \mathbb{E}(\hat{\sigma}^2) = \sigma^2$$

Propiedades estadísticas OLS

Bajo los supuestos de Gauss, se tiene las siguientes **propiedades estadísticas**:

1. Linealidad: los LSE son una función lineal de la v.a y.
2. Insesgadez:
$$E(\hat{\beta}_j) = \beta_j, \quad \forall j = 1, 2 \tag{6}$$
3. Eficiencia: varianza mínima.

Linealidad LSE

$$
\begin{aligned}
\hat{\beta}_2 &= \frac{\sum(x_{i2}-\bar{x}_2)(y_i-\bar{y})}{\sum(x_{i2}-\bar{x}_2)^2}; \quad \bar{y} = \frac{1}{n}\sum y_j \\
&= \frac{\sum(x_{i2}-\bar{x}_2)\left(y_i - \frac{1}{n}\sum y_j\right)}{\sum(x_{i2}-\bar{x}_2)^2} \\
&= \frac{\sum(x_{i2}-\bar{x}_2)y_i - \frac{1}{n}\sum(x_{i2}-\bar{x}_2)\sum y_j}{\sum(x_{i2}-\bar{x}_2)^2}; \quad \sum(x_{i2}-\bar{x}_2) = 0 \\
&= \frac{\sum(x_{i2}-\bar{x}_2)y_i}{\sum(x_{i2}-\bar{x}_2)^2} \\
&= \sum\left(\frac{x_{i2}-\bar{x}_2}{\sum(x_{j2}-\bar{x}_2)^2}\right)y_i \\
&= \sum a_i y_i
\end{aligned}
$$

Insesgamiento LSE

$$\hat{\beta}_2 = \frac{\sum(x_{i2} - \bar{x}_2)(y_i - \bar{y})}{\sum(x_{i2} - \bar{x}_2)^2}; \quad y_i - \bar{y} = \beta_2(x_{i2} - \bar{x}_2) + (u_i - \bar{u})$$

$$= \frac{\sum(x_{i2} - \bar{x}_2)[\beta_2(x_{i2} - \bar{x}_2) + (u_i - \bar{u})]}{\sum(x_{i2} - \bar{x}_2)^2}$$

$$= \frac{\beta_2 \sum(x_{i2} - \bar{x}_2)^2 + \sum(x_{i2} - \bar{x}_2)(u_i - \bar{u})}{\sum(x_{i2} - \bar{x}_2)^2}; \quad \sum(x_{i2} - \bar{x}_2) = 0$$

$$= \beta_2 + \frac{\sum(x_{i2} - \bar{x}_2)u_i}{\sum(x_{i2} - \bar{x}_2)^2}$$

$$\mathbb{E}(\hat{\beta}_2) = \beta_2 + \frac{\sum(x_{i2} - \bar{x}_2)\mathbb{E}(u_i)}{\sum(x_{i2} - \bar{x}_2)^2}; \quad \mathbb{E}(u_i) = 0$$

$$\mathbb{E}(\hat{\beta}_2) = \beta_2$$

Eficiencia LSE

$$
\begin{aligned}
var(\hat{\beta}_2) &= var\left(\frac{\sum(x_{i2}-\bar{x}_2)u_i}{\sum(x_{i2}-\bar{x}_2)^2}\right) \quad [de\ (5)] \\
&= \frac{1}{\left[\sum(x_{i2}-\bar{x}_2)^2\right]^2} var\left(\sum(x_{i2}-\bar{x}_2)u_i\right); \quad var\left(\sum(x_{i2}-\bar{x}_2)u_i\right) = \sigma^2 \sum(x_{i2}-\bar{x}_2)^2 \\
&= \frac{\sigma^2}{\sum(x_{i2}-\bar{x}_2)^2} \\
\tilde{\beta}_2 &= \sum c_i y_i \\
\mathbb{E}(\tilde{\beta}_2) = \beta_2 &\Rightarrow \sum c_i = 0, \quad \sum c_i x_i = 1 \\
var(\tilde{\beta}_2) &= \sigma^2 \sum c_i^2 \\
\sum c_i^2 &\geq \frac{1}{\sum(x_{i2}-\bar{x}_2)^2} \\
var(\tilde{\beta}_2) &\geq var(\hat{\beta}_2)
\end{aligned}
$$

Tabla ANOVA

El análisis de varianza (ANOVA) en *regression analysis* comprende la descomposición de la varianza de y:

$$SCT = SCE + SCR \tag{7}$$

Se sabe que

$$(y_i - \bar{y}) = (\hat{y}_i - \bar{y}) + (\hat{u}_i) \tag{8}$$

Al elevar al cuadrado y aplicar sumatorio,

$$\begin{aligned} \sum (y_i - \bar{y})^2 &= \sum [(\hat{y}_i - \bar{y}) + (\hat{u}_i)]^2 \\ &= \sum [(\hat{y}_i - \bar{y})^2 + 2(\hat{y}_i - \bar{y})\hat{u}_i + \hat{u}_i^2] \\ &= \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{u}_i^2 \end{aligned} \tag{9}$$

Tabla ANOVA

Gráficamente:

Bondad de ajuste

1. **Coeficiente de determinación** (muestral):

$$R^2 = \frac{SCE}{SCT} \tag{10}$$

2. **Root mean squared error** (RMSE) o error estándar de regresión:

$$ee = \hat{\sigma}_{\hat{u}}, \quad \hat{\sigma}^2_{\hat{u}} = \frac{\sum \hat{u}_i^2}{n-2}$$

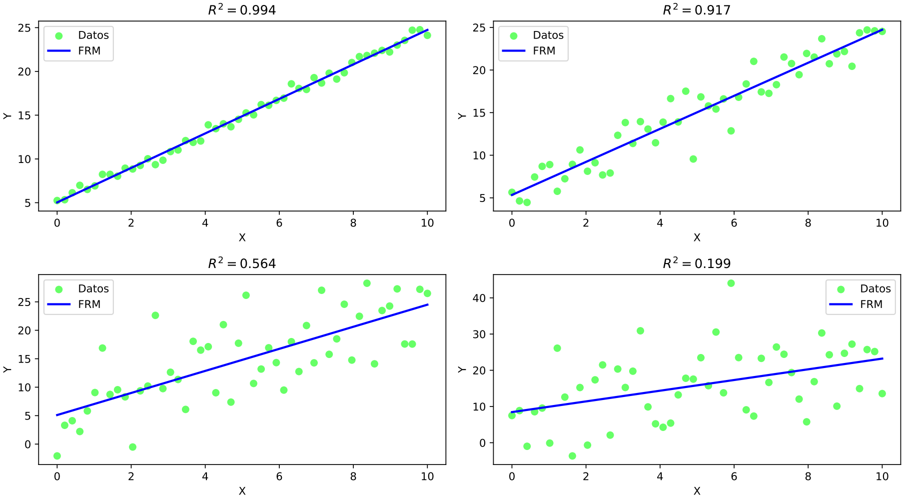
donde $\hat{\sigma}^2_{\hat{u}}$ es conocido como **mean squared error** (MSE).

3. **Mean absolute error** (MAE):

$$MAE = \frac{1}{n} \sum |\hat{u}_i| \tag{11}$$

DAT

Bondad de ajuste



Bondad de ajuste

Ejemplo 1

A partir de los datos del fichero metria02.csv estimar la ecuación de demanda de camotes piuranos. Los precios están expresados en soles/kg y las cantidades en kg. Hallar la varianza de los LSE y los indicadores de bondad de ajuste. Interpretar cada resultado.

Referencias I

Goldberger, A. (1991). *A course in econometrics*. Harvard University Press.

Greene, W. (2018). *Econometric Analysis*. Pearson, 8th edition.